Heat Transfer

MEP 212

Dr. Tamer Maher
Assistant Professor
Mechanical Power Eng. Dep.

References

- Incropera and DeWitt. “Fundamentals of Heat and Mass Transfer” sixth edition Wiley, 2006.
- Yunus A. Cengel “Heat transfer a practical approach”, 2nd edition.
- Holman, J. P., “Heat Transfer” McGraw-Hill Book Co., 1990.

Introduction

Heat Transfer – What?

- Heat is thermal energy (J)
- Heat transfer is the science of thermal energy transfer due to ***temperature difference.*** (W)

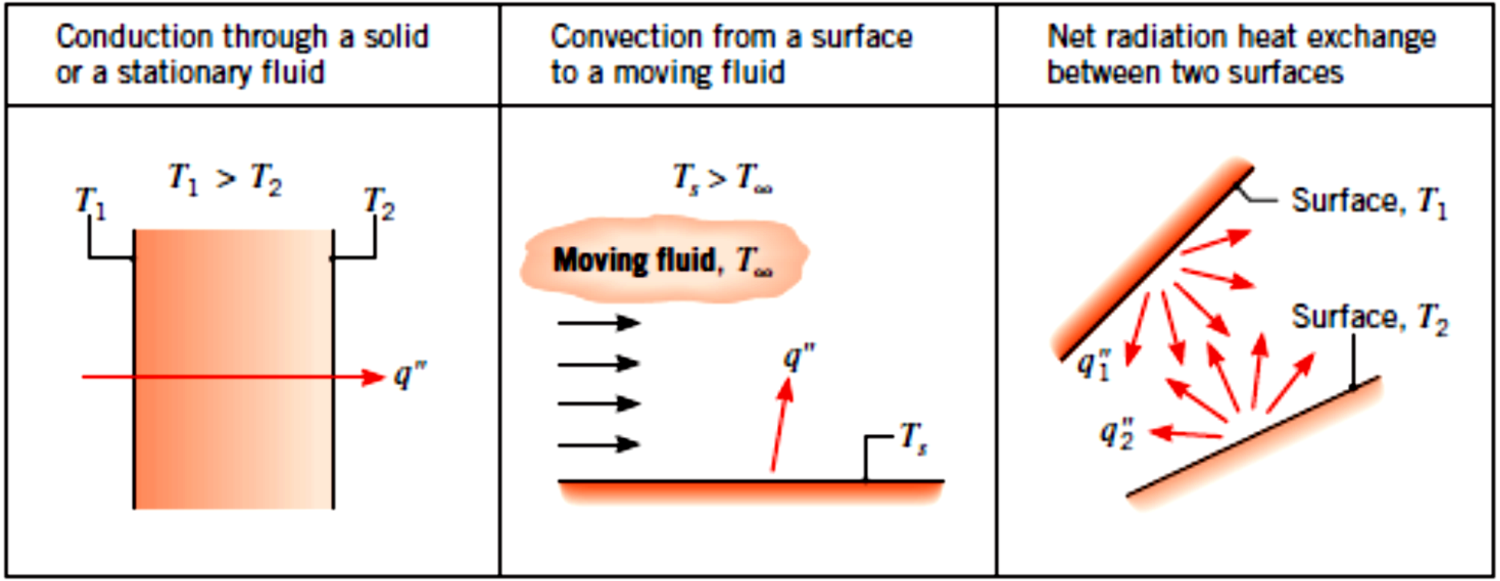
- **Driving force** is the temperature difference

As the voltage difference for the electrical current

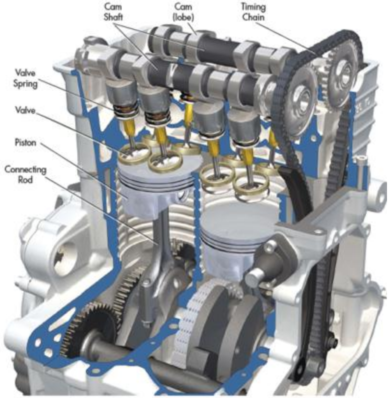
As the pressure drop for the fluid flow

Heat Transfer – How?

- There are 3 heat transfer modes:



Heat Transfer – applications

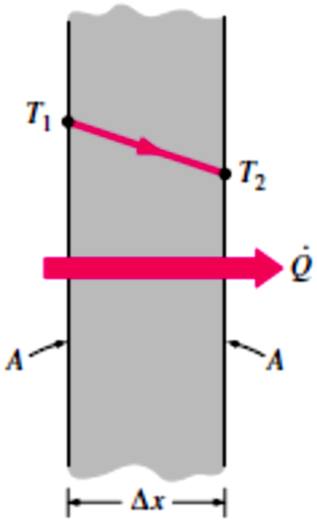


Conduction

the transfer of energy from the more energetic to the less energetic particles of a substance due to interactions between the particles.

In solids, it is due to the combination of *vibrations* of the molecules in a lattice and the energy transport by *free electrons.*

In gases and liquids, conduction is due to the *collisions* and *diffusion* of the molecules during their random motion.

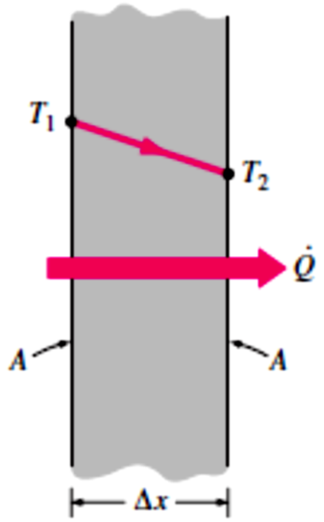


Conduction-Fourier’s law of heat conduction

$$\text{Rate of heat conduction} \propto \frac{(\text{Area})(\text{Temperature difference})}{\text{Thickness}}$$

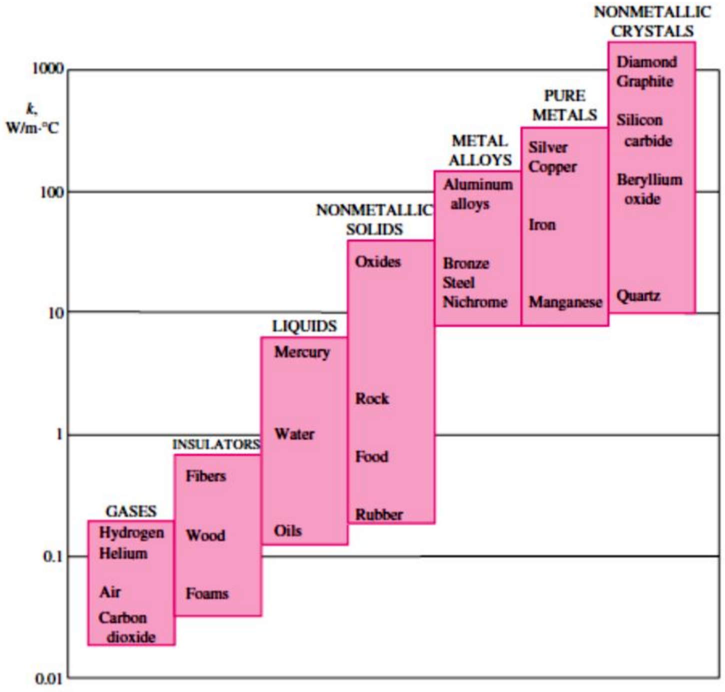
$$\dot{Q}_{\text{cond}} = -kA\frac{dT}{dx} \qquad \text{(W)}$$

$$\dot{Q}_{\text{cond}} = kA\frac{T_1 - T_2}{\Delta x} = -kA\frac{\Delta T}{\Delta x} \qquad \text{(W)}$$



Conduction-Thermal conductivity

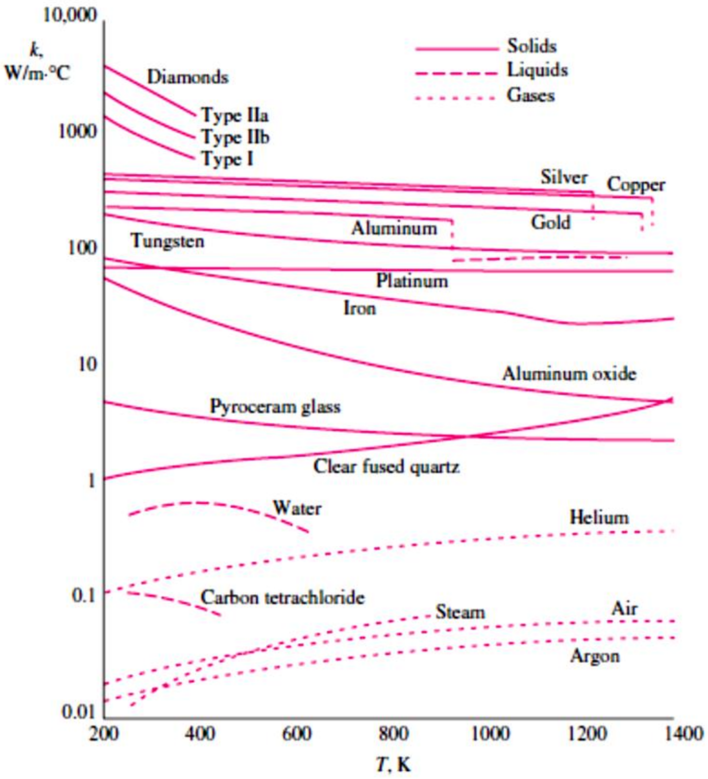
The thermal conductivity k (W/m · °C)is a measure of a material's ability to conduct heat.



The thermal conductivities of some materials at room temperature

Material	k, W/m · °C*
Diamond	2300
Silver	429
Copper	401
Gold	317
Aluminum	237
Iron	80.2
Mercury (l)	8.54
Glass	0.78
Brick	0.72
Water (l)	0.613
Human skin	0.37
Wood (oak)	0.17
Helium (g)	0.152
Soft rubber	0.13
Glass fiber	0.043
Air (g)	0.026
Urethane, rigid foam	0.026

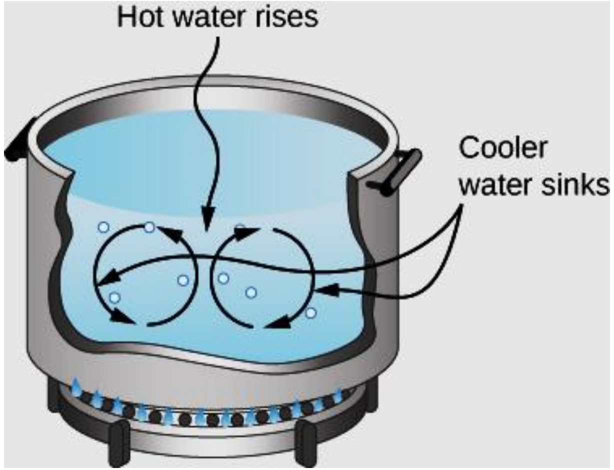
Conduction-Thermal conductivity

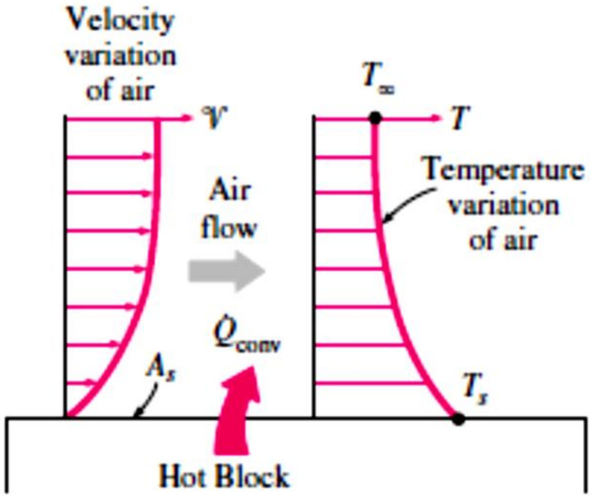


Convection

- **Convection** is the mode of energy transfer between a solid surface and the adjacent liquid or gas that is in motion, and it involves the combined effects of *conduction* and *fluid motion.*
- The faster the fluid motion, the greater the convection heat transfer. In the absence of any bulk fluid motion, heat transfer between a solid surface and the adjacent fluid is by pure conduction.
- The presence of bulk motion of the fluid enhances the heat transfer between the solid surface and the fluid, but it also complicates the determination of heat transfer rates.

Convection


Hot water rises
Cooler
water sinks


Velocity
variation
of air
$\mathcal{V}$
T_∞
T
Temperature
variation
of air
Air
flow
$\dot{Q}_{conv}$
A_s
T_s
Hot Block

Convection-Newton’s law of cooling

$$\dot{Q}_{conv} = hA_s(T_s - T_\infty) \qquad \text{(W)}$$

where *h* is the ***convection heat transfer coefficient*** in $W/m^2 \cdot °C$

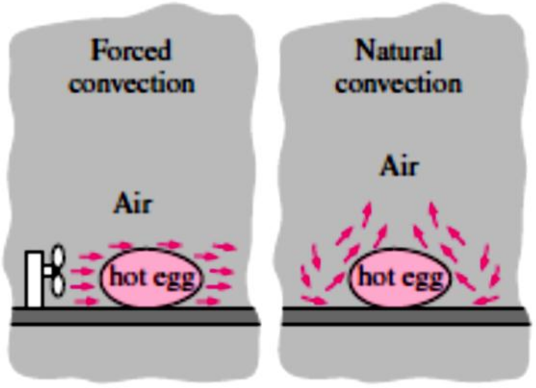
The convection heat transfer coefficient *h* is not a property of the fluid. It is an experimentally determined parameter whose value depends on all the variables influencing convection such as:

- The surface geometry
- Type of fluid
- The nature of fluid motion,
- The properties of the fluid,
- and the bulk fluid velocity.

Convection-types

Typical values of convection heat transfer coefficient

Type of convection	h, W/m^2 · °C*
Free convection of gases	2–25
Free convection of liquids	10–1000
Forced convection of gases	25–250
Forced convection of liquids	50–20,000
Boiling and condensation	2500–100,000



Radiation

- **Radiation** is the energy emitted by matter in the form of *electromagnetic waves* (or *photons*) as a result of the changes in the electronic configurations of the atoms or molecules.
- Unlike conduction and convection, the transfer of energy by radiation does not require the presence of an *intervening medium.*
- All bodies at a temperature above absolute zero emit thermal radiation.
- Radiation is a *volumetric phenomenon,* and all solids, liquids, and gases emit, absorb, or transmit radiation to varying degrees.

Radiation

- However, radiation is usually considered to be a *surface phenomenon* for solids that are opaque to thermal radiation such as metals, wood, and rocks since the radiation emitted by the interior regions of such material can never reach the surface, and the radiation incident on such bodies is usually absorbed within a few microns from the surface.

Radiation- Black body radiation

$$\dot{Q}_{\text{emit, max}} = \sigma A_s T_s^4 \qquad \text{(W)}$$

where $\sigma = 5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4$ *Stefan–Boltzmann constant.*

The idealized surface that emits radiation at this maximum rate is called a **blackbody,** and the radiation emitted by a blackbody is called **blackbody radiation**

Radiation- Real body radiation

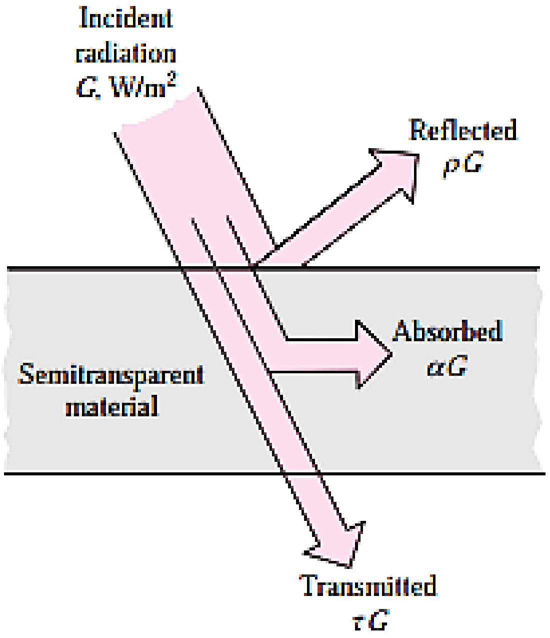
The radiation emitted by all real surfaces is less than the radiation emitted by a blackbody at the same temperature, and is expressed as

$$\dot{Q}_{emit} = \varepsilon \sigma A_s T_s^4 \qquad \text{(W)}$$

where ε is the **emissivity** of the surface. The property emissivity, whose value is in the range 0 1, is a measure of how closely a surface approximates a blackbody for which ε=1.

Radiation- Real body radiation

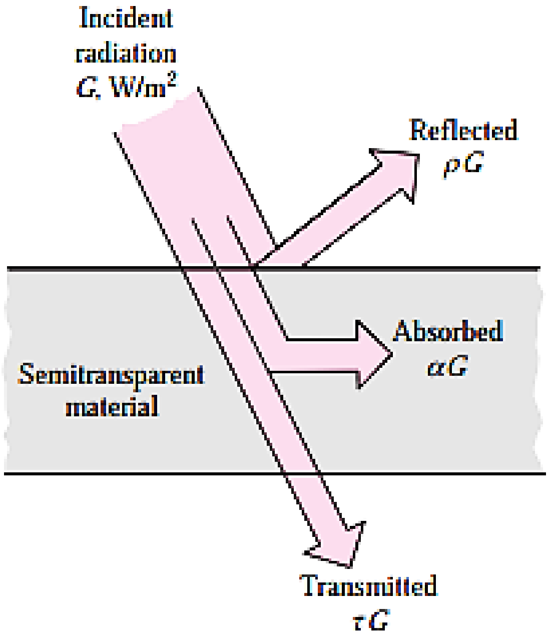
Another important radiation properties of a surface is its **absorptivity α**, **reflectivity ρ** and **transmissivity τ**. Absorptivity which is the fraction of the radiation energy incident on a surface that is absorbed by the surface. Reflectivity which is the fraction of the radiation energy incident on a surface that is reflected by the surface. Transmissivity which is the fraction of the radiation energy incident on a surface that is transmitted by the surface.



Radiation- Real body radiation

Like emissivity, its value is in the range $0 \le \alpha \le 1$, $0 \le \rho \le 1$ and $0 \le \tau \le 1$ .A blackbody absorbs the entire radiation incident on it. That is, a blackbody is a perfect absorber ($\alpha = 1$) as it is a perfect emitter.

In general, both ε, α, ρ and τ of a surface depend on the temperature and the wavelength of the radiation.



Radiation- Real body radiation

$$Q_a + Q_r + Q_t = Q_0$$

$$\frac{Q_a}{Q_0} + \frac{Q_r}{Q_0} + \frac{Q_t}{Q_0} = 1$$

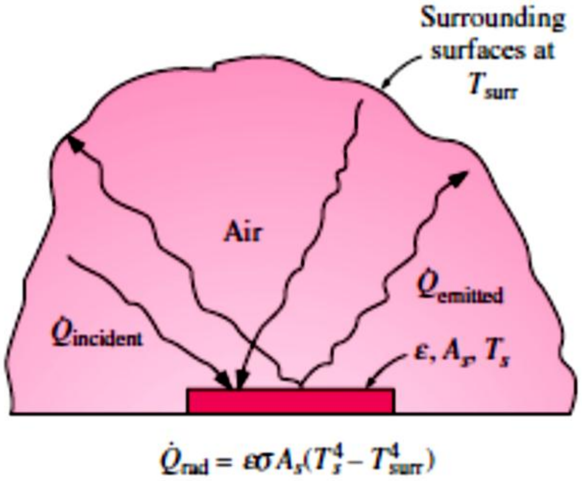
$$\alpha + \rho + \tau = 1$$

α	absorptivity
ρ	reflectivity
τ	transmissivity

Radiation- Real body radiation

Kirchhoff's law of radiation states that the emissivity and the absorptivity of a surface at a given temperature and wavelength are equal.

$$\dot{Q}_{rad} = \varepsilon\sigma A_s (T_s^4 - T_{surr}^4) \qquad \text{(W)}$$



Thank you